

## CAPSULE: PROJECT - 2

### Loan Eligibility Prediction – Documentation

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#### **1. Introduction**

Loan eligibility prediction is a machine learning problem used to predict whether a loan application will be approved or rejected based on the applicant's information. In this project, the loan dataset was explored and prepared by checking the data, handling missing values, converting categorical data into numerical values, and scaling numerical features. The focus was on data preprocessing, feature engineering, and pipeline implementation rather than building a final prediction model. **Two approaches were used:**

a) Notebook performs the preprocessing steps manually using Pandas and Scikit-learn, b) Notebook uses a Scikit-learn Pipeline with ColumnTransformer to organize and apply the preprocessing steps in a consistent way.

## 2. Objective

**The main objective** of this project was to understand and implement the pre processing steps required to prepare a loan eligibility dataset for machine learning. The work focused on inspecting the dataset, understanding the different types of features, identifying missing values and duplicate records, analyzing numerical relationships, checking the distribution and skewness of numerical variables, and transforming the data into a format that can be used by machine learning algorithms.

**Another important objective** was to understand the difference between performing feature engineering manually and implementing the same pre processing operations through a machine learning pipeline. The project therefore included missing-value imputation, categorical feature encoding using One-Hot Encoding, and numerical feature scaling using Min-Max Scaling. The final pipeline combined these operations into a sequential workflow so that the pre processing steps could be executed together.

## 3. Dataset Review

The dataset used for the project contains 614 records and 13 original columns, including Loan\_ID, applicant demographic information, financial information, property information, and the target variable Loan\_Status. After removing Loan\_ID, the working dataset contained 614 rows and 12 columns. The features include Gender, Married, Dependents, Education, Self\_Employed, ApplicantIncome, CoapplicantIncome, LoanAmount, Loan\_Amount\_Term, Credit\_History, and Property\_Area, while Loan\_Status represents the loan approval outcome.

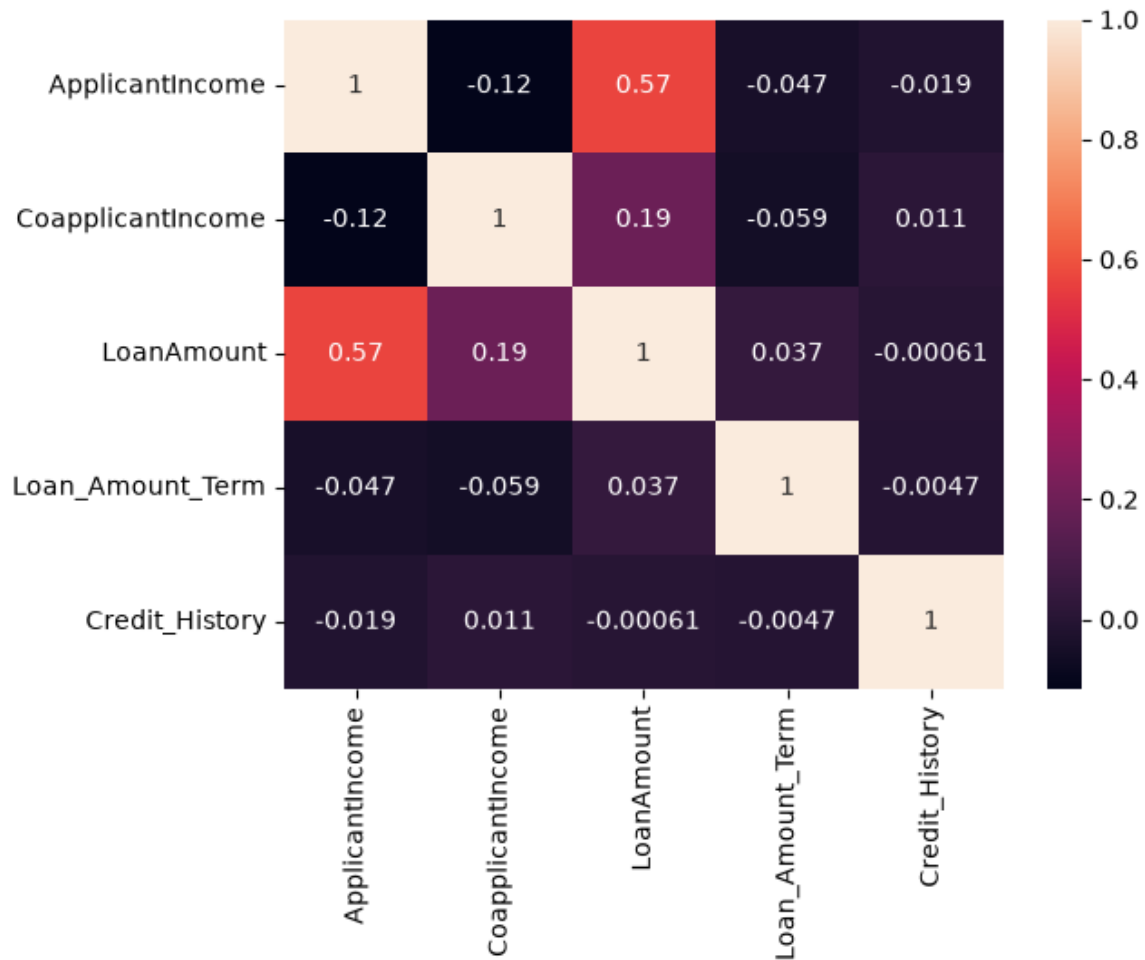
**The initial dataset review** showed that both numerical and categorical variables were present. The numerical variables were ApplicantIncome, CoapplicantIncome, LoanAmount, Loan\_Amount\_Term, and Credit\_History. The categorical variables included Gender, Married, Dependents, Education, Self\_Employed, Property\_Area, and Loan\_Status. Missing values were found in several columns: Gender had 13 missing values, Married had 3, Dependents had 15, Self\_Employed had 32, LoanAmount had 22, Loan\_Amount\_Term had 14, and Credit\_History had 50. No duplicate rows were found in the dataset.

**The descriptive analysis** also showed considerable differences in the ranges of the numerical features. For example, ApplicantIncome ranged from 150 to 81,000, while CoapplicantIncome ranged from 0 to 41,667 and LoanAmount ranged from 9 to 700. The median values were 3,812.5 for applicant income, 1,188.5 for coapplicant income, and 128 for loan amount.

**The categorical analysis** showed that Gender contained Male and Female values, Married contained Yes and No, Dependents contained 0, 1, 2 and 3+, Education contained Graduate and Not Graduate, and Property\_Area contained Urban, Rural and Semiurban. The target variable Loan\_Status contained two categories, Y and N, with Y being the more frequent class in the dataset.

### Fig 1 – Correlation Heatmap

The correlation analysis was performed on the five numerical features. The strongest positive relationship observed was between ApplicantIncome and LoanAmount, with a correlation of approximately 0.565. CoapplicantIncome and LoanAmount had a weaker positive correlation of approximately 0.189. The remaining numerical relationships were relatively weak. This analysis was useful for understanding the relationships between numerical variables before preprocessing.



## 4. Procedure

### ❖ Data Loading and Initial Inspection

The dataset was first loaded using Pandas' `read_csv()` function. After loading, the first few records were examined using `df.head()` and the dataset structure was checked using `df.shape`, `df.columns`, and `df.info()`. The initial inspection confirmed that the dataset contained 614 records and a mixture of numerical and categorical features. `Loan_ID` was removed from the working dataset during the feature engineering process because it functions as an identifier rather than a meaningful predictive feature.

The data types and non-null counts were then examined. This step helped identify which variables were numerical and which were categorical, as well as where missing values existed. The dataset contained five numerical variables and seven categorical/string variables after the identifier was removed. This classification was important because different preprocessing methods were required for numerical and categorical data.

### ❖ Missing Value Analysis and Imputation

Missing values were identified using `df.isna().sum()`. Missing data was present in both numerical and categorical features. For the numerical variables `LoanAmount`, `Loan_Amount_Term`, and `Credit_History`, missing values were filled using the median value of each respective column. Median imputation was used to provide a representative central value without being as strongly affected by extreme values as the mean.

For the categorical variables Gender, Married, Dependents, and Self\_Employed, missing values were replaced using the mode, or most frequently occurring value. After the imputation process, the dataset was checked again using `df.isna().sum()`, and all columns showed zero missing values in the manual feature-engineering notebook.

### ❖ Checking Categorical Values

The unique values of the categorical columns were examined to understand the categories available in each feature. For example, Gender contained Male and Female, Dependents contained 0, 1, 2 and 3+, while Property\_Area contained Urban, Rural and Semiurban. This step helped determine which categorical variables needed to be converted into numerical representation before being passed to a machine learning algorithm.

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### ❖ Checking Duplicate Records

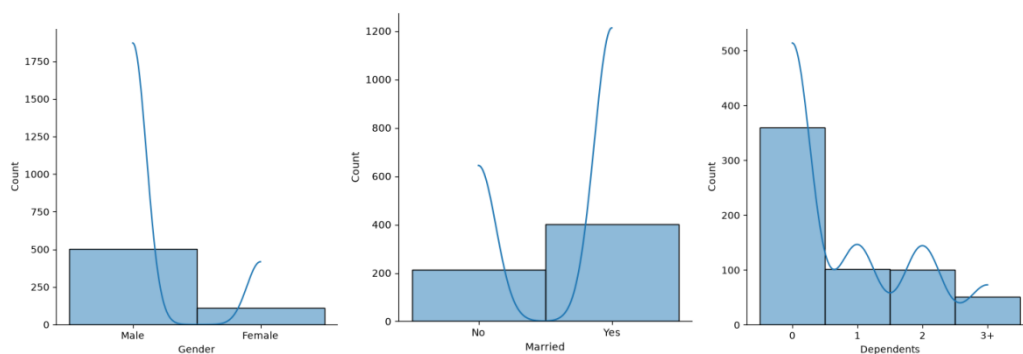
Duplicate records were checked using `df.duplicated().sum()`. The result was **0**, meaning that no duplicate rows were identified in the dataset. Therefore, no rows were removed as part of duplicate-record cleaning.

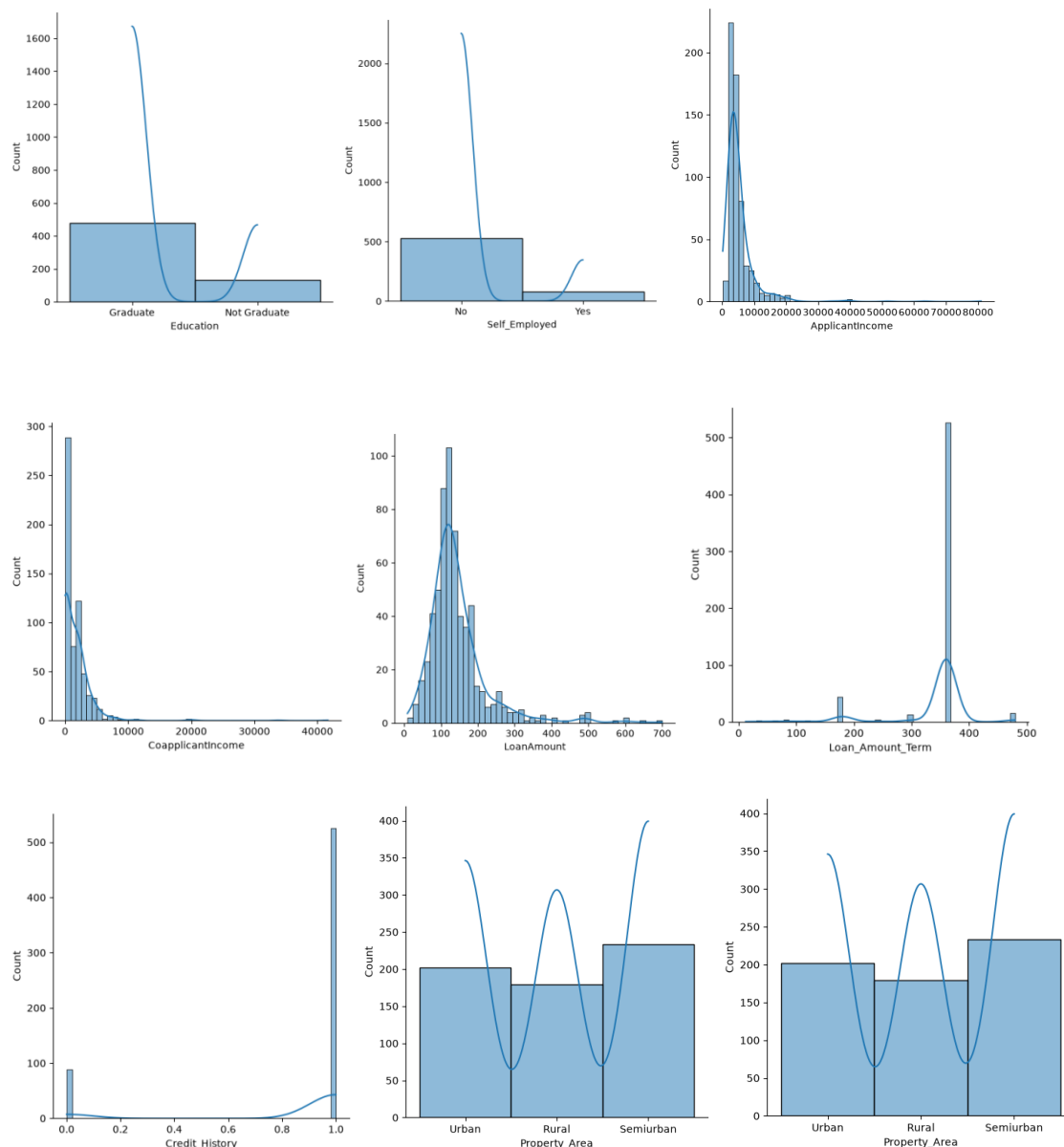
### ❖ Skewness Analysis

The distribution of the numerical variables was examined using skewness values and distribution plots. The analysis showed that several numerical variables were highly skewed. ApplicantIncome had a skewness of approximately 6.54, CoapplicantIncome approximately 7.49, and LoanAmount approximately 2.74. Loan\_Amount\_Term and Credit\_History showed negative skewness of approximately -2.40 and -2.02, respectively.

Distribution plots were also generated for the dataset columns to visually inspect their distributions. This analysis helped identify that the numerical variables did not all follow a similar distribution and that some contained strong skewness. The skewness analysis was therefore used as part of understanding the numerical features before applying scaling.

**Figure 2: Distribution plots used to examine the numerical features.**





## ❖ One-Hot Encoding

Categorical features cannot generally be directly used by machine learning algorithms in their original text form. Therefore, One-Hot Encoding was explored to convert categorical values into numerical indicator columns.

In the Feature Engineering notebook, OneHotEncoder was first applied to Gender and Dependents. For example, the Gender column was converted into Gender\_Female and Gender\_Male, while Dependents was converted into separate indicator columns such as Dependents\_0, Dependents\_1, Dependents\_2, and Dependents\_3+. After creating these encoded columns, the original Gender and Dependents columns were removed.

The Pipeline notebook expanded this approach by applying One-Hot Encoding to Gender, Married, Dependents, Education, Self\_Employed, and Property\_Area. `handle_unknown="ignore"` was used so that an unseen category during transformation would not cause the transformation process to fail. The encoded output was configured as dense data using `sparse_output=False`.

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## ❖ Numerical Feature Scaling

After the categorical transformation, numerical features were scaled using `MinMaxScaler`. The numerical columns selected for scaling were:

- `ApplicantIncome`
- `CoapplicantIncome`
- `LoanAmount`
- `Loan_Amount_Term`
- `Credit_History`

Min-Max scaling transforms the numerical values into a common range, normally between 0 and 1. This prevents variables with larger numerical ranges, such as income, from having a disproportionately large numerical scale compared with variables such as credit history. In the manual feature-engineering notebook, the scaler was fitted to these five numerical columns and the transformed values replaced the original values.

For example, the resulting transformed dataset shows `ApplicantIncome` values such as approximately 0.070489 instead of the original value 5849. Similarly, `LoanAmount` and `Loan_Amount_Term` were represented on the scaled numerical range.

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## ❖ Pipeline Implementation

The second notebook implemented the preprocessing operations using a Scikit-learn Pipeline. Four transformation stages were created: `trf1`, `trf2`, `trf3`, and `trf4`. The purpose of dividing the preprocessing into separate transformers was to handle different types of preprocessing operations in a controlled sequence.

### **TRF1 – Numerical Imputation:**

`trf1` used `SimpleImputer(strategy="median")` on `LoanAmount`, `Loan_Amount_Term`, and `Credit_History`. This ensured that missing numerical values were replaced with the corresponding median values. The remaining columns were passed through using `remainder="passthrough"`.

### **TRF2 – Categorical Imputation:**

`trf2` used `SimpleImputer(strategy="most_frequent")` on `Gender`, `Married`, and `Self_Employed`. The most frequently occurring category was used to replace missing values in these columns.

### **TRF3 – One-Hot Encoding:**

`trf3` used `OneHotEncoder` on `Gender`, `Married`, `Dependents`, `Education`, `Self_Employed`, and `Property_Area`. The encoder generated separate binary columns for the categories of these variables. `handle_unknown="ignore"` was included to make the transformation safer when new categorical values appear.

### **TRF4 – Numerical Scaling:**

`trf4` applied `MinMaxScaler` to `ApplicantIncome`, `CoapplicantIncome`, `LoanAmount`, `Loan_Amount_Term`, and `Credit_History`. This converted the selected numerical features to a common scale.

Finally, all four transformers were combined sequentially:

**TRF1 → TRF2 → TRF3 → TRF4**

The complete pipeline was then executed using `pipe.fit_transform(df)`. This allowed all preprocessing stages to be applied in the defined order without manually executing every transformation separately.

## 5. Result

The feature-engineering process successfully transformed the original loan dataset into a more machine-learning-ready form. Missing numerical values were handled using median imputation, while categorical missing values were handled using the most frequent category in the manual feature-engineering process. After preprocessing, the manual notebook confirmed that all columns contained zero missing values.

The categorical transformation converted text-based information into numerical binary features. In the manual feature-engineering process, Gender and Dependents were converted into separate binary columns. The resulting dataset contained features such as Gender\_Female, Gender\_Male, Dependents\_0, Dependents\_1, Dependents\_2, and Dependents\_3+. This allowed categorical information to be represented numerically while retaining the information contained in the original categories.

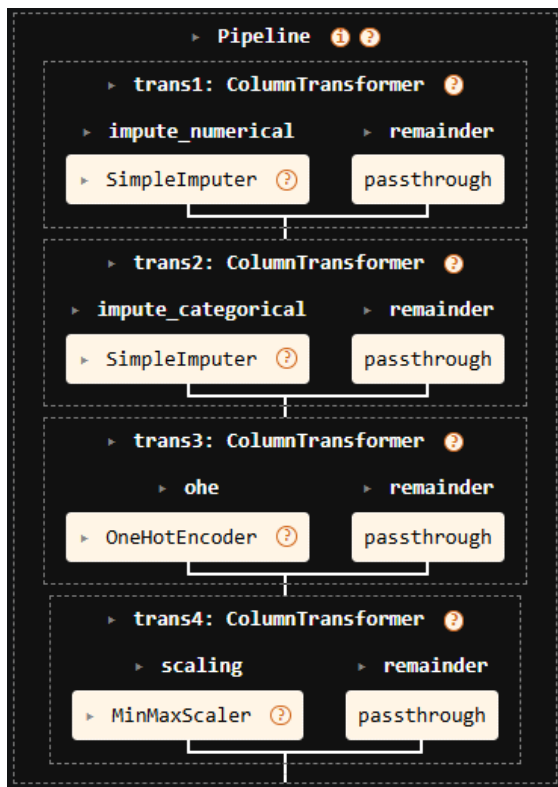
The numerical features were also successfully scaled using Min-Max Scaling. The transformed values were placed on a common scale, making the numerical features more comparable in terms of magnitude. For example, the original ApplicantIncome values were transformed into values between the Min-Max range, and the same transformation was applied to CoapplicantIncome, LoanAmount, Loan\_Amount\_Term, and Credit\_History.

### Pipeline Result

The pipeline produced a transformed dataset with **614 rows and 23 columns**. The output contained the scaled numerical features followed by the generated one-hot encoded categorical features. Examples of the resulting columns include Gender\_Female, Gender\_Male, Married\_No, Married\_Yes, Dependents\_0, Dependents\_1, Dependents\_2, Dependents\_3+, Education\_Graduate, Education\_Not Graduate, Self\_Employed\_No, Self\_Employed\_Yes, and the different Property\_Area categories.

The transformed output demonstrates that the preprocessing pipeline successfully combined numerical scaling and categorical encoding into one workflow. The first five numerical features in the transformed output are represented by scaled values, while the categorical information is represented by binary 0/1 values. For example, a male applicant is represented by Gender\_Male = 1 and Gender\_Female = 0, while the applicant's dependent category is represented through the corresponding dependent indicator column.

**Figure 3: Pipeline Structure**



## **6. Conclusion**

This project provided a practical understanding of how raw data can be prepared for machine learning through feature engineering and preprocessing. The loan eligibility dataset contained both numerical and categorical variables as well as missing values, so different preprocessing techniques were required. The analysis identified missing values, checked duplicate records, examined categorical values, analyzed numerical distributions and skewness, and investigated correlations between numerical features. These steps helped establish a clear understanding of the dataset before transformation.

The project also demonstrated the advantage of using a Scikit-learn pipeline to organize preprocessing operations. Median imputation, most-frequent categorical imputation, One-Hot Encoding, and Min-Max Scaling were combined into a single sequential workflow. The final pipeline produced a transformed dataset of 614 rows and 23 columns, making the data more suitable for subsequent machine learning tasks. The work therefore established the preprocessing stage required before training and evaluating a loan eligibility prediction model.

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